

Governance & Constitution

Compliance Audit â€” 2026-08-07

Revision: 1 **Last modified:** 2026-08-07T19:10:54Z **Status:** active **Scope:** Full-repository audit against the inherited Helix Constitution (constitution/CLAUDE.md, constitution/Constitution.md) + verification of docs/REMAINING_WORK_PLAN.md (2026-06-14) closure claims + live-toolchain health. Assembled from 6 parallel read-only subagent audits, 2026-08-07. No file was modified by the audit itself; every finding below cites file:line, commit, or command output as evidence (Â§11.4.6 â€” no guessing).

Baseline: HEAD 0d05ec1 on main. docs/CONTINUATION.md was last updated at commit 646b295 (2026-06-16, â€œSession 14â€”) â€” 21 commits / ~52 days behind HEAD at audit time.

0. How to execute this plan

Same discipline as REMAINING_WORK_PLAN.md: each GA-NN item is TDD-first (RED before fix, Â§11.4.43/Â§11.4.115), gets a regression guard where it closes a defect (Â§11.4.135), and closes only with pasted live command output (no self-certification words without evidence). Items marked **OPERATOR-DECISION** are surfaced, not auto-executed. Non-contending items run in parallel subagent streams (Â§11.4.70/Â§11.4.103).

Phase 0 â€” Toolchain unblock (P0, prerequisite for verifying everything else)

GA-01 â€” Python test toolchain is ABI-broken; no project .venv

- **Evidence:** live run of the documented command `python3 -m pytest tests/unit/ -q --import-mode=importlib` crashes during plugin auto-load: `ModuleNotFoundError: No module named 'rpds.rpds' / No module named 'pydantic_core.pydantic_core'`. `python3` resolves to 3.14.6; every native extension in `~/.local/lib/python3/site-packages` is compiled for `cpython-313`. No `.venv/` exists in the repo. **Zero tests can currently run** â€” CONTINUATION.mdâ€™s â€œ4277 passedâ€” claim is unreproducible today.
- **Blast radius:** blocks the entire unit suite, RW-21 verification, and 3 Challenge scripts (`download_proxy_deep_challenge.sh`, `private_tracker_html_challenge.sh`, `search_deep_coverage_challenge.sh`) that need `.venv`.
- **Fix:** create a project-pinned `.venv` (Python 3.12, matching `pyproject.toml`â€™s `target-version/mypy python_version`), `pip`

install -r tests/requirements.txt (+ any download-proxy requirements), verify pytest tests/unit/ -q --import-mode=importlib actually collects and runs, paste the real N-passed/N-failed count. Document the exact bootstrap in docs/TESTING.md and CONTINUATION.md™s Quick-Start (which currently omits venv activation entirely).

- **Acceptance:** real pytest run completes (not crashes) with a captured summary line; count compared against the 4277 baseline and any delta explained.
- **Priority: P0.**

GA-02 “ Frontend Vitest toolchain broken (stale node_modules)

- **Evidence:** cd frontend && npx vitest run fails before any test executes: Error: Cannot find module '@analogjs/vite-plugin-angular' while loading vitest.config.ts. frontend/node_modules exists (332M) but is incomplete/stale.
- **Fix:** cd frontend && npm ci (or npm install if no lockfile match), re-run npx vitest run, capture the real pass count.
- **Priority: P0.**

GA-03 “ Extension Vitest toolchain never bootstrapped

- **Evidence:** extension/node_modules absent entirely. All 8 challenges/extension/*.sh scripts fail cleanly and honestly (FAIL: vitest not installed ... run 'cd extension && npm install' first) rather than false-passing “ the challenges themselves are not buggy, the environment is just never set up. Blocks verification of the “œ379 passed / 33 spec files“ claim in CONTINUATION.md Session 10.
- **Fix:** cd extension && npm install, re-run npx vitest run + all 8 extension challenges, capture real counts.
- **Priority: P1.**

Phase 1 “ Governance/tracking seam reconnection

Root cause: the mandated tracking chain (CONTINUATION.md “ Issues.md/Fixed.md/workable_items.db “ README entry-point) stopped being touched on 2026-06-16 while real engineering work (19 substantive commits) kept landing cleanly and correctly on main. The code is fine; the seam that is supposed to keep governance docs synced with code was silently disconnected “ precisely the “œprose exists, nobody wired the seam“ failure mode the constitution™s own Â§11.4.205/Â§11.4.226/Â§11.4.227 anchors describe.

GA-04 " CONTINUATION.md 52-day / 21-commit staleness gap

- **Evidence:** `git log --oneline 646b295..HEAD` = 21 commits (19 substantive), spanning 2026-06-16T23:30 to 2026-08-07T21:52, none reflected in the file. Themed: Cyrillic/UTF-8 encoding-fix wave (6 commits, 029c045..cdec407), tracker/plugin fixes (7), egress/proxy/Jackett feature wave (4: 2f160cd,4e62eca,b44a2e4,be5062d), doc-sync (1), host-mirror sync (2, no-op for CONTINUATION purposes).
- **Violates:** 11.10 / 11.4.131 (standing session-resumption file, always current).
- **Fix:** author a new "Session 15" entry summarizing all 19 substantive commits by theme, update `**Last commit:**`/`**Last modified:**`/`**Revision:**`, reconcile the "In flight" NEXT ITEM section (the P0 Cyrillic item it names as in-flight is now DONE per commit 4aee3c4/8918d4b " verify and close it in the doc).
- **Priority:** P0.

GA-05 " Lava P1-P4 porting work never landed as tracked workable items

- **Evidence:** docs/PORTING-FROM-LAVA.md (created be5062d, 2026-07-01) maps 4 items to Lava-audit finding numbers: P1 5 durable remote-exec, P2 0/4 egress/VPN routing, P3 3 configurable outbound proxy, P4 8 Jackett cookie-session login. All four are implemented in code (scripts/deploy-remote.sh, containers/pkg/remote/executor.go, scripts/egress-via-vpn.sh, download-proxy/src/config/proxy.py, qBitTorrent-go/internal/httpx/proxy.go, commit 2f160cd explicitly "port Lava 8 finding" but `grep -in "lava|egress|BOBA_UPSTREAM_PROXY|durable-run|remote-exec" docs/Issues.md docs/Fixed.md` returns zero hits. A 6/9 item ("anti-bluff QA verdict/diagnostics strengthening") is listed applicable in the porting doc's own triage table but no commit closes it " status unknown.
- **Violates:** 11.4.148 / 11.4.202 / 11.4.208 (every actionable finding must land as a tracked item, not merely a commit message).
- **Fix:** create BOB-064..067 via the workable-items tool (not raw MD edits, to keep DB "i_ZMD custody intact " workable-items validate/diff already confirmed the DB "i_ZMD pair is internally in sync, just frozen at BOB-063), cite the implementing commit + file as evidence, close as Implemented/Fixed. Determine the true status of the 6/9 item and either close it with evidence or track it honestly as open.
- **Priority:** P0.

GA-06 " docs/PORTING-FROM-LAVA.md missing 11.4.44 revision header

- **Evidence:** `grep -n Revision docs/PORTING-FROM-LAVA.md` empty.
- **Fix:** add `**Revision:**` N / `**Last modified:**` header block.
- **Priority:** P2.

GA-07 “ README.md has no §11.4.57/§11.4.212 doc-link entry-point section

- **Evidence:** README.md’s ## Documentation section (lines 108–162) links 20+ docs but contains **zero** references to docs/Issues.md, docs/Fixed.md, docs/CONTINUATION.md, docs/PORTING-FROM-LAVA.md, docs/browser_extension/Status.md, docs/browser_extension/RELEASE_READINESS.md, docs/COMPLETION_STATUS.md, or docs/RELEASE_READINESS_20260616.md. Every tracker/status doc audited here is an orphan relative to README.
- **Violates:** §11.4.57 (Tracked-Items + Status Documents mandated section) / §11.4.212 (README as canonical entry point for ALL project documentation).
- **Fix:** add the mandated table (Document | Last modified | Revision | Markdown | HTML | PDF) covering every tracker/status doc in the repo, including this audit doc and REMAINING_WORK_PLAN.md.
- **Priority: P0** (structural, repo-wide).

GA-08 “ docs/browser_extension/Status.md stale by 4 commits

- **Evidence:** Rev 15, 2026-06-13T13:10:00Z. Commits since, untouched in the doc: a410b91 (coverage), 0897248 (video-confirm), ee4a01b (crash-hardening), 0dc247b (qBittorrent 5.x compat).
- **Fix:** update Status.md + regenerate Status_Summary.md + HTML/PDF siblings (§11.4.45/§11.4.56).
- **Priority: P2.**

GA-09 “ workable_items.db: BOB-008 missing operator_block_details row

- **Evidence:** workable-items validate --db docs/workable_items.db at 1 violation: “BOB-008: Operator-blocked with no operator_block_details row (§11.4.148 D3)”. The prose in Issues.md DOES contain the “Operator-Block-Details:” line “this is a DB-column population gap, not a doc gap.”
- **Fix:** populate via the workable-items tool’s proper subcommand (not raw SQL).
- **Priority: P3.**

GA-10 “ No successor to docs/RELEASE_READINESS_20260616.md; no top-level v1.0.0 ledger

- **Evidence:** the doc is a self-labeled point-in-time snapshot (HEAD 8b9ef90); 20 commits since have touched the audited surfaces. Its own text names “no top-level proxy/merge-service v1.0.0 readiness ledger” as an open blocker “this is RW-20 in the existing plan, still NOT-DONE (confirmed by the RW-plan verification agent).

- **Fix:** create the ledger doc (dedupes with RW-20 " track under one ticket, not two).
 - **Priority:** P2.
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Phase 2 " Security correctness gaps (corrections to RW-plan closure claims)

GA-11 " RW-02 incomplete: 2 mutating routes still unauthenticated

- **Evidence:** routes.py/scheduler.py's auth rollout covers download/upload/file/magnet (routes.py) and POST/DELETE schedules " but PATCH /api/v1/schedules/{id} (scheduler.py:108) and PUT /api/v1/theme (routes.py:82) have **no** Depends(require_api_token). Confirmed by omission: tests/security/test_hooks_schedules_auth.py:194-198's own _MUTATING enumeration never included them " the original fix and its test were scoped incompletely together.
- **Fix (TDD):** RED " extend _MUTATING to include PATCH schedule + PUT theme, confirm the new assertions fail (401 expected, currently unauthenticated) " add Depends(require_api_token) to both routes " GREEN " live curl-verify 401-unauth / 200-with-token on a running container.
- **Priority:** P0 (security).

GA-12 " RW-06 unverifiable: rutracker ReDoS fix not confirmed deployed

- **Evidence:** source fix present (plugins/rutracker.py:140, {0,512} bound); no live container running on the audit host to check config/qBittorrent/nova3/engines/rutracker.py (the installed copy) or measure live parse timing " a "11.4.108 SOURCE" ARTIFACT/RUNTIME gap that is unverified, not necessarily unfixed.
- **Fix:** bring up the compose stack (./start.sh), run ./install-plugin.sh, verify grep '{0,512}' config/qBittorrent/nova3/engines/rutracker.py on the container, capture timing of a large rutracker result page.
- **Priority:** P1 (needs live stack).

GA-13 " RW-07 partial: nnmclub e2e SKIP-on-404 fallback never removed

- **Evidence:** tests/e2e/test_live_stack_evidence.py:265 still contains the skip fallback; the acceptance criterion ("œun-skip" is unmet regardless of live container state " the route existing in source (download-proxy/src/api/auth.py:350) isn't what was asked for.

- **Fix:** with the live stack up, confirm `curl :7187/api/v1/auth/nmclub/status` returns 200, remove the skip fallback, let the assertion run for real, capture the pass.
 - **Priority: P1** (needs live stack).
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Phase 3 – Test-type integrity (constitution’s –non-unit tests hit real services– rule)

GA-14, GA-15, GA-16 – Three test files mislabeled by directory mock the exact system under test

- **Evidence:**
 - `tests/integration/test_merge_api.py` (577 lines) – own docstring: “Uses FastAPI TestClient with mocked SearchOrchestrator.” Every route test `@patch("api.routes._get_orchestrator")`.
 - `tests/e2e/test_full_pipeline.py` – docstring claims “End-to-end”, but patches `SearchOrchestrator._get_enabled_trackers/_search_tracker` with a synthetic fake – no real tracker HTTP call ever happens.
 - `tests/contract/test_tracker_stats_contract.py:80-81` – same monkeypatch.setattr pattern on the same two methods.
- **Violates:** CLAUDE.md’s inherited rule: “Mocks, stubs are permitted ONLY in unit tests | All other test types MUST interact with real fully implemented System” (A\$11.4.27).
- **Fix:** for each file – (a) relocate the mocked version into `tests/unit/` under an honest name (the coverage IS valuable as unit-level route-contract testing), AND (b) author a new real-service replacement in the original directory using the project’s EXISTING live-stack fixtures (`tests/fixtures/compose.py`, `tests/fixtures/services.py`, `tests/integration/test_fixtures_bring_up_services.py` already demonstrate this pattern is viable in this repo) so the directory’s contract (`integration/e2e/contract` = real services) is actually met.
- **Priority: P1** (each file independent, parallelizable).

GA-17 – #legacy-untriaged skip-tag family never actually triaged

- **Evidence:** ~10 sites (`tests/e2e/test_public_trackers_return_results.py:128`, `tests/unit/test_openapi_frozen.py:15,23`, `tests/unit/test_plugin_smoke.py:45`, `tests/docs/test_no_broken_links.py:51`, `tests/fixtures/services.py:9`, others) carry `SKIP-OK: #legacy-untriaged` – syntactically satisfies the format, but the “ticket” is a literal placeholder admitting it was never triaged.

- **Fix:** triage each site individually â€” determine the real reason, either fix/un-skip or replace the placeholder with a real ticket id + reason.
- **Priority:** P3.

GA-18 â€” 7 permanently-skipped dead Jackett-autoconfig Python tests

- **Evidence:** tests/contract/test_jackett_autoconfig_contract.py:18,24, tests/integration/test_jackett_autoconfig_real.py:71,77,83, tests/e2e/test_jackett_autoconfig_e2e.py:32,38, tests/security/test_jackett_autoconfig_secrets.py:41 â€” all skip with â€œendpoint moved to boba-jackett:7189â€”. Migration confirmed deliberate: download-proxy/src/api/__init__.py:301-306 explicitly removes the Python endpoint in favour of the Go service (RW-13 evidence).
 - **Fix (Â§11.4.124 investigate-before-remove â€” investigation above already satisfies the proof requirement):** delete these dead test bodies in one dedicated commit citing the git-history/migration evidence; the replacement coverage lives in qBitTorrent-go/tests/integration/jackett_db_test.go and the Go autoconfig package.
 - **Priority:** P3.
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Phase 4 â€” Go backend parity (blocked on operator decision)

GA-19 â€” OPERATOR-DECISION: is --profile go parity still a release goal?

- **Evidence:** this is RW-09, unresolved since 2026-06-14. docs/migration/PARITY_GAPS.md and docs/features/Status.md:52,194 are unrevised (â€œLast audited 2026-04-27â€”, 6 ported / 4 partial / 8 missing). Confirmed-missing since the plan was written: scheduler driver loop (qBitTorrent-go scheduler files have no time.Ticker), metadata enricher (no equivalent package), public-tracker plugin fan-out (no exec.Command fan-out in internal/service). Jackett autoconfig (RW-13) is now PARTIAL â€” implemented as its own Go service, canonical per download-proxy/src/api/__init__.py:301-306, but BEP48/scrape validation and a CAPTCHA REST endpoint are still absent from the Go side.
- **This gates:** RW-10 (scheduler driver â€” â€œhighest-impact silent functional hole on the Go pathâ€” per the RW-planâ€™s own words), RW-11 (enricher), RW-12 (plugin fan-out), the remainder of RW-13, and GA-20â€™s stress/chaos items RW-14/RW-15 to the extent theyâ€™re Go-path-specific.
- **Decision needed:** commit real scope to close the 3 missing Go subsystems (multi-day effort), or mark the Go backend a documented future blueprint and stop implying itâ€™s a near-term parity target in Status.md.

- **Priority: P1, surfaced via Â§11.4.66 â€” not auto-executed.**

GA-20 â€” RW-14/RW-15 stress/chaos coverage claims were inaccurate; still NOT-DONE

- **Evidence:** commit a410b91 (â€œclose Go + extension coverage gaps RW-14/RW-16â€”â€” actually added Go internal/jackett/internal/logging unit tests and an extension parser test â€” NOT Â§11.4.85-conformant stress/chaos coverage for the tracker-fetch/cookie-auth path (RW-14) or the scheduler+hooks+SSE-broker surface (RW-15). No matching test files exist. RW-16â€™s Go-DB concurrent-write test (jackett_db_test.go:440-500) predates the plan (commit 8936545, 2026-04-27) and has no fault-injection â€” genuinely PARTIAL, not the chaos coverage the plan calls for.
 - **Fix:** author real tests/stress/test_tracker_fetch_stress_chaos.py (download-proxy cookie-auth path, overlapping the RW-03 SSRF surface) and tests/stress/test_scheduler_hooks_sse_stress_chaos.py, plus a chaos-fault-injection extension to the existing Go DB concurrency test (process-kill / resource-exhaustion, not just concurrent writes). Correct the historical claim in a follow-up doc note (Â§11.4.6 â€” donâ€™t leave the false â€œclosedâ€” claim standing).
 - **Priority: P2.**
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Phase 5 â€” Challenges / HelixQA infrastructure bugs

GA-21 â€” jackett_autoconfig_clean_slate.sh polls a retired endpoint

- **Evidence:** 3 references to /api/v1/jackett/autoconfig/last (lines 8, 58, 62) â€” this endpoint was deliberately removed per CLAUDE.md (â€œThe Python /api/v1/jackett/autoconfig/last endpoint was removedâ€”â€”). The challenge would always fail step 5 against the current architecture.
- **Fix:** update to poll the Go :7189 autoconfig-runs API (GET /api/v1/jackett/autoconfig/runs or /runs/{id} per qBitTorrent-go/internal/jackettapi/router.go).
- **Priority: P1.**

GA-22 â€” All 6 HelixQA bank symlinks are dangling absolute macOS paths

- **Evidence:** challenges/helixqa-banks/*.yaml â†’ symlinks to /Volumes/T7/Projects/Boba/submodules/helixqa/banks/*.yaml, authored on a different dev machine. Broken on this (and any non-that-exact-Mac) host. The real files exist and are correct at submodules/

helixqa/banks/*.yaml (37-entry populated submodule, confirmed not empty).

- **Fix:** replace with relative symlinks (`ln -sf ../../submodules/helixqa/banks/<name>.yaml`).
- **Priority: P1.**

GA-23 “ HelixQA bank action steps hardcode /Volumes/T7/Projects/Boba/...

- **Evidence:** e.g. `boba-boba-ctl.yaml: cd /Volumes/T7/Projects/Boba/cmd/boba-ctl && go build ...; boba-docs-chain.yaml: bash /Volumes/T7/Projects/Boba/scripts/docs_chain.sh`. Non-portable to this or any other checkout.
- **Fix:** parametrize with a `$PROJECT_ROOT/$REPO_ROOT` variable the HelixQA runner resolves at invocation time, or make paths script-relative.
- **Priority: P2.**

GA-24 “ no_suspend_calls_challenge.sh false-positives on a governance doc

- **Evidence:** flags `constitution/docs/scripts/guard-forbidden-commands.md` because that doc *quotes* the forbidden `systemctl/loginctl` verbs it documents/enforces. Its `EXCLUDE_PATHS` allowlist covers `CONSTITUTION.md/AGENTS.md/CLAUDE.md` but not this file “ a carrier false-positive, the exact `11.4.201` pattern the constitution itself warns about.
- **Fix:** add the file to `EXCLUDE_PATHS` in `scripts/host-power-management/check-no-suspend-calls.sh`.
- **Priority: P3.**

Phase 6 “ CONST-033 real signal (HIGH “ project’s own hard-ban class)

GA-25 “ host_no_auto_poweroff_challenge.sh found 1 genuine post-fix poweroff broadcast

- **Evidence:** 4/5 checks pass; 1 fails “ a “œwill power off“ journal broadcast recorded AFTER the documented fix marker (2026-04-28T20:54:31+03:00).
- **This is the exact incident class CLAUDE.md’s own CONST-033 Operational Note exists for.** Per that note, the correct response is triage BEFORE assuming host-code causation and BEFORE dismissing it: (1) uptime cross-check against any alleged downtime window, (2) `journalctl -k --since "24 hours ago" | grep -iE "will suspend|systemd-suspend"`, (3) `journalctl -k --since "24 hours ago" | grep -iE "oom-kill|killed process"` + decode the `oom_memcg cgroup` path if present, (4) re-run both CONST-033

challenges, (5) document findings in docs/incidents/<date>-* .md regardless of outcome. **Only read-only diagnostic commands** â€” nothing that triggers a power-state transition, per the absolute Hard Ban.

- **Priority: P0** â€” triage this first, independent of everything else in this plan, given the projectâ€™s own stated severity for this incident class (prior data-loss history cited in CLAUDE.md).
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Phase 7 â€” Gate-scoping bug

GA-26 â€” ruff check . sweeps submodules/containers/ (a separately-governed submodule)

- **Evidence:** [tool.ruff] in pyproject.toml has no exclude/extend-exclude at all. 71 ruff violations found; sampled violations are exclusively in submodules/containers/scripts/resource-policy/*.py â€” a submodule that owns its own engineering standards per Â§11.4.28 decoupling. This inflates bobaâ€™s own lint-gate count with debt that isnâ€™t bobaâ€™s to fix.
 - **Fix:** add extend-exclude = ["submodules/", "constitution/", ".worktrees/"] to [tool.ruff], re-run ruff check ., report the TRUE boba-only violation count (may be 0, may be nonzero â€” verify, donâ€™t assume).
 - **Priority: P2.**
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Phase 8 â€” Governance-layer ambiguity (needs operator clarification)

GA-27 â€” OPERATOR-DECISION: Hard Stop #3 vs.Â this projectâ€™s documented maintenance workflow

- **Evidence:** the inherited constitutionâ€™s Hard Stop #3 states container orchestration must be owned exclusively by the projectâ€™s own binary/orchestrator, with direct docker/podman commands â€œprohibited as workflows.â€” This projectâ€™s own CLAUDE.md â€œCritical Constraintsâ€” section explicitly prescribes direct podman exec qbittorrent-proxy ..., podman restart qbittorrent-proxy, and podman compose down && podman compose up -d as the mandated procedure for Python-source reload, plugin reload, and compose/env changes.
- **Decision needed:** either (a) formally classify this as an intentional, documented exception per Â§11.4.17 (with a one-line rationale â€” e.g.Â â€œrestart-level granularity below full stack lifecycle is exempt because start.sh/stop.sh donâ€™t yet expose itâ€”), or (b) extend start.sh/stop.sh with subcommands that wrap these exact operations (e.g. ./start.sh --restart-proxy, ./start.sh --reload-plugins)

so operators never type raw podman commands, closing the gap for real.

- **Priority: P1, surfaced via Â§11.4.66 â€” not auto-executed.**

Summary table

Phase	Items	Blocking on live infra?	Blocking on operator decision?
0 â€” Toolchain unblock	GA-01..03	No	No
1 â€” Governance/tracking	GA-04..10	No	No
2 â€” Security corrections	GA-11..13	GA-12/13 yes	No
3 â€” Test-type integrity	GA-14..18	GA-14..16 partial (real-service replacements)	No
4 â€” Go parity	GA-19..20	No	GA-19 yes
5 â€” Challenge/HelixQA bugs	GA-21..24	No	No
6 â€” CONST-033 signal	GA-25	No (read-only triage)	No
7 â€” Gate scoping	GA-26	No	No
8 â€” Governance ambiguity	GA-27	No	GA-27 yes

Cross-reference: GA-11..13, GA-19..20 correct/extend docs/REMAINING_WORK_PLAN.mdâ€™s RW-02/06/07/09/14/15 entries â€” that documentâ€™s status table should be updated in the same pass to avoid two divergent trackers describing the same defects (Â§11.4.186 anti-divergence).

Definition of done for this audit: every GA item above reaches DONE (with live-captured evidence), OPERATOR-DECISION items (GA-19, GA-27) are resolved, REMAINING_WORK_PLAN.mdâ€™s corrected items are reconciled, README carries the doc-link section, CONTINUATION.md is current, and every fix ships with a regression guard + (where the surface is user-facing) a HelixQA Challenge entry.